

The Future Beyond LLMs

Exploring Agentic AI

Raphaël Semeteys

WORLDLINE

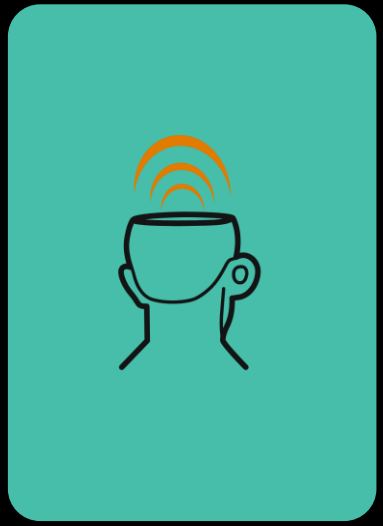
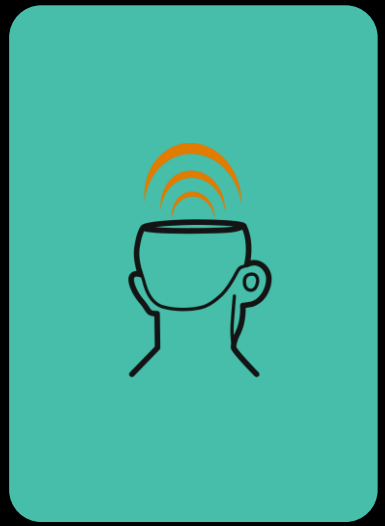
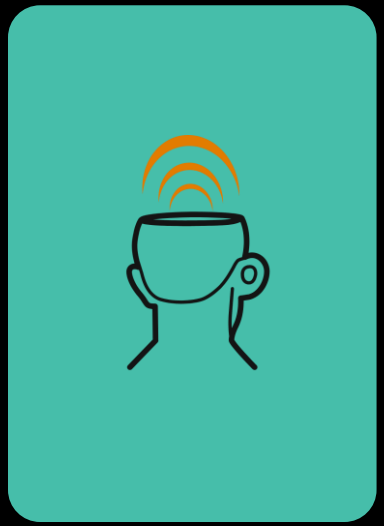
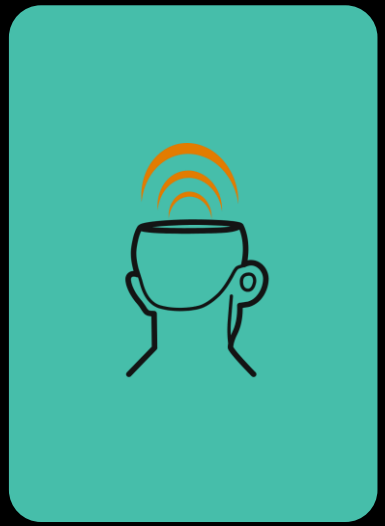
3 October 2024

DEVOXX MOROCCO

DEVOXX MOROCCO

You want to know the future of Generative AI?

Draw four cards!

WORLDLINE

Raphaël Semeteys

Head of TechSquad
Senior Architect, Open Source Expert

TechForum eXplore 2024

- Since 1999
- Based in Paris, France
- raphiki.github.io



Get in touch

@RaphaelSemeteys

dev.to/raphiki/

linkedin.com/in/raphaelsemeteys/

WORLDLINE

DEVOXX MOROCCO

Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 43+ billion
transactions per year



€250M spent in R&D
every year



Handling 150+
payment methods

WORLDLINE

Yann LeCun, Chief AI Scientist at Meta



My prediction is that no sane person will be using ***LLMs*** *maybe in 5 years. They* ***will be outdated****.*

Understanding the real world, having persistent memory, reasoning, and planning*: these are 4 characteristics of intelligent behavior that LLMs basically cannot do, or can only do in a very superficial and approximate manner.*

OpenAI’s 5-step plan for AGI

Artificial General Intelligence is a form of AI that can perform **better than humans across every task**

OpenAI Imagines Our AI Future

Stages of Artificial Intelligence

Level 1	Chatbots, AI with conversational language	**Now**: OpenAI GPT
Level 2	Reasoners, human-level problem solving	**Preview**: OpenAI o1
Level 3	Agents, systems that can take actions	**Tomorrow**
Level 4	Innovators, AI that can aid in invention	**Before 10 years**
Level 5	Organizations, AI that can do the work of an organization	

Source: Bloomberg reporting




DEVOXX MOROCCO
Before the doom of AGI... let's focus on Agents
WORLDLINE

What are Agents?

Software entities that can perform tasks on behalf of humans

Goal Setting

Predefined objectives that can be static or dynamically adjusted based on current state and environment

State Estimation

Actual state of the environment is estimated based on sensors and memory while handling uncertainties, noise, and incomplete information



Planning

Dynamic generation of a sequence of actions that will lead to achieving the goal

Perception

Sensory information is processed and used to build an internal representation of the current state of the environment

Action

Selection of the best action based on plan and current state. Execution of the action and potential change in the state of environment

Where do we stand today on the road to Level 5 Agents?

- Perception
- State Estimation
- Goal Setting
- Planning
- Action



Evolving
Large Language Models



Blooming
Agent Frameworks



Emerging
Large Action Models

Large Language Models

DEVOXX MOROCCO

First 2 levels of GenAI (conversation, reasoning)

Goal Setting

One shot prompting
Reflection with Chain of Thought

State Estimation

Internal representation (embeddings)
Contextual Augmentation



Planning

No real workflow generation

Perception

NLP, Vision, Audio
Multimodality

Action

Generation of text, image, sound
Tool calling

WORLDLINE

Large Language Models – Tool Calling



> Calling HTTP endpoint

> Response received

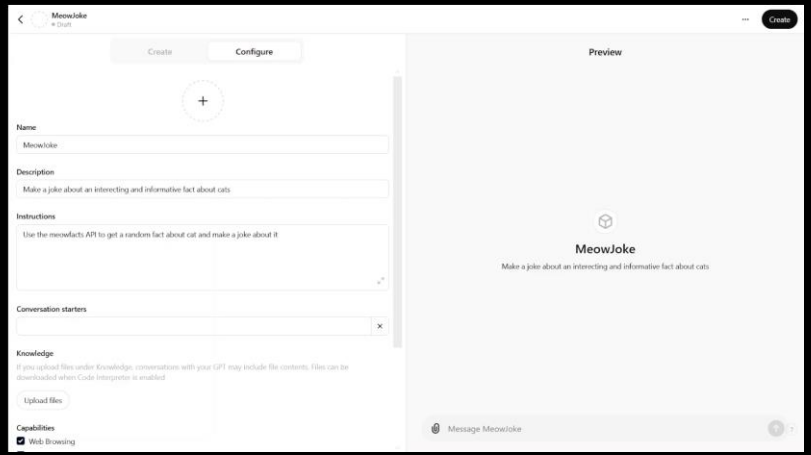
Connect LLMs to external tools and systems

- Empowering a LLM with external capabilities or resources
- Decision to use a Tool based on its description and the general

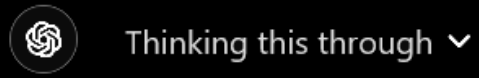
- Features
  - Built-in Tools (code interpreter, search)
  - Third-party plugins (Browser, Expedia, OpenTable…)
  - Function calls
  - Assistants (Custom GPTs) with API calls and memory (Threa

- Demo: tell a joke on cats

- Next: Gorilla paper (Retrieval Aware Training)
  - Reason with API constraints by training
  - Adapt to API changes at test time



Large Language Models – Chain of Thought

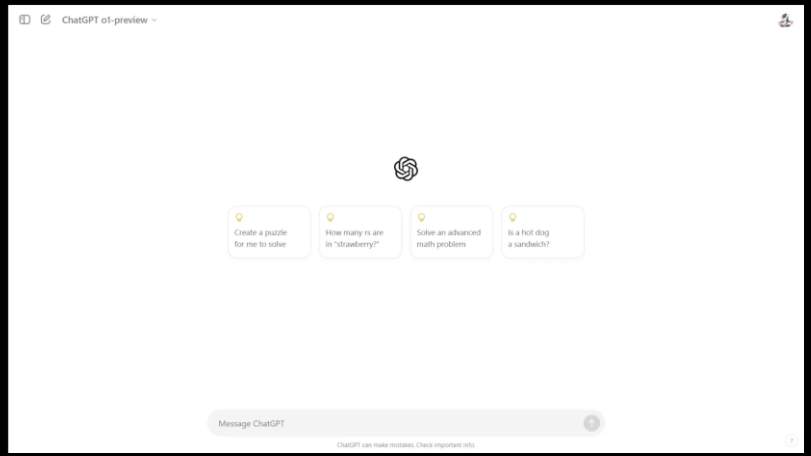


OpenAI o1: new reflection and reasoning capacities

- New model family released by OpenAI on September 12
- Preview versions: o1 and o1-mini

- Reasoning with Chain of Thought
  - Refine strategies, recognize and correct mistakes
  - Detailed Chain of Thought hidden to the user
  - Notion of reasoning tokens

- Demo: solve a complex math problem



Agent Frameworks

From Prompt Engineering to Flow Engineering

Goal Setting

Task-orientation
Beyond one-shot prompting

State Estimation

Notion of State or Memory



Planning

Multi-agent architectures
Notion of workflow

Perception

Use of LLM

Action

Use of LLMs
Tool calling
Human-in-the-loop

Agent Frameworks - LangGraph

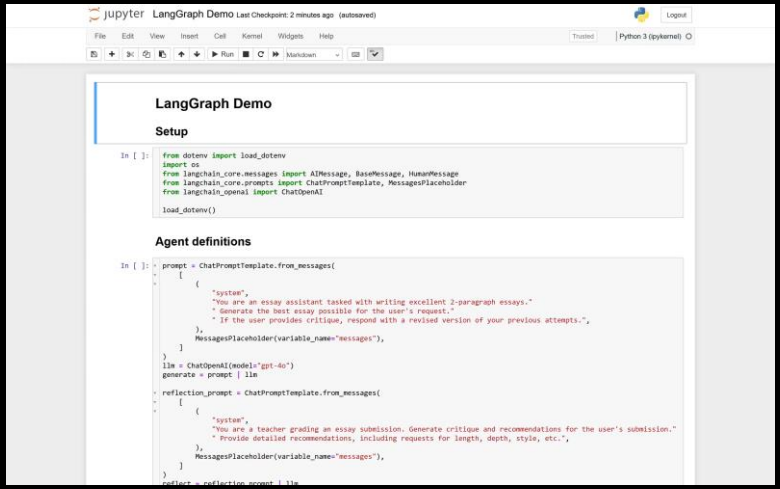


Build resilient language agents as graphs

- Open source project (MIT license)
- Led by LangChain since December 2023

- Library to create agent and multi-agent workflows as graphs
- Features
  - State management and graph-based workflows
  - Tool calling
  - Human-in-the loop

- Demo: generate and reflect (two agents)



Agent Frameworks - Autogen

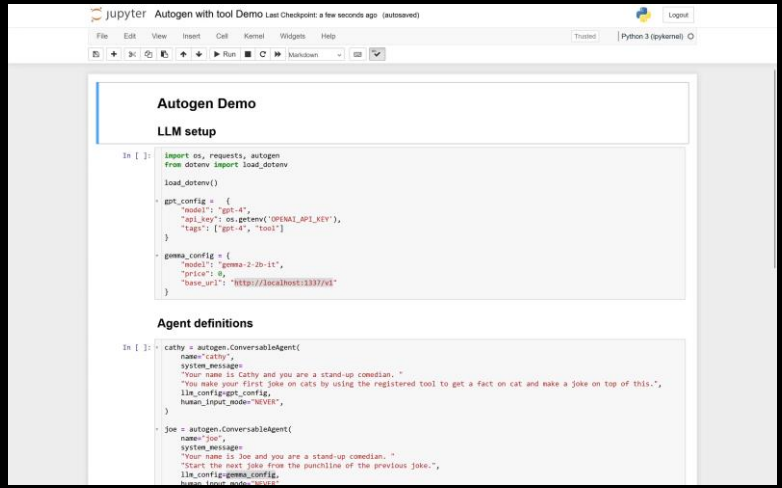


A programming framework for agentic AI

- Open source project (MIT license)
- Led by Microsoft Research since May 2023

- Autogen focuses on conversational interactions between agents
- Features
  - Customized agents (role, prompt, persona, capacities)
  - Code execution
  - Human-in-the-loop

- Demo: conversation between stand-up comedians (2 agents)



Agent Frameworks – CrewAI

DEVOXX MOROCCO

crewai

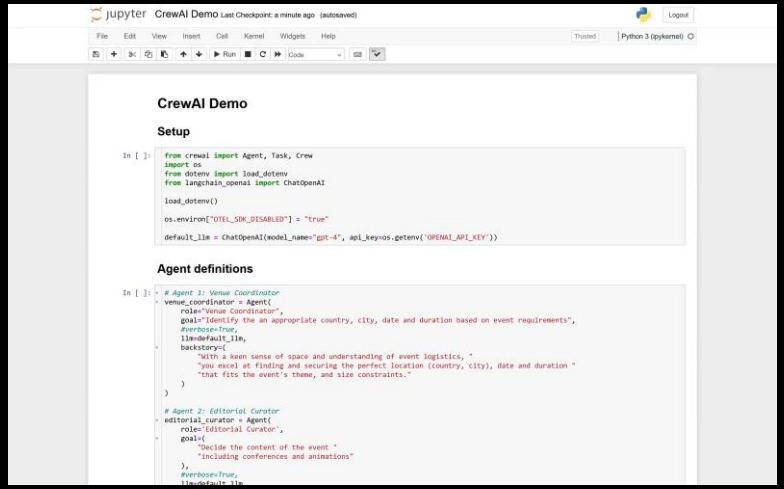
Framework for orchestrating role-playing, autonomous AI agents

- Open source project (MIT license)
- Led by CrewAI Inc. since December 2023

- Task orientation and distribution between a crew of agents
- Features
  - Role-based agents orchestrated by a manager
  - Tool calling
  - Human-int-the-loop

- Demo: organizing a Tech event (3 agents)

- A step toward Organizational AI
  - CrewAI is still explicit
  - Next: Meta Agents (Automated Design of Agentic Systems)



WORLDLINE

Agent Frameworks – Comparison

The 3 frameworks offers similar general features to build multi-agent architectures

The differ on how workflows are defined and managed

		AG	crewai
Structure	Directed graph	Dynamic conversational flow	Hierarchical team structure
Workflow Type	Pre-defined graph	Collaborative task solving	Hierarchical task distribution
State Management	Explicit	Implicit through conversation history	Task-based
Agent Interaction	Defined by graph	Free-form conversations	Role-based interaction

Large Action Models

DEVOXX MOROCCO

Actions & Behavior as a language

Goal Setting

Trained on domain-specific goals
and action-oriented Datasets

State Estimation

Internal representation
Symbols and logical relationships



Planning

Complex and specific
sequences of actions

Perception

Use of LLM
Specific sensors

Action

Observe, learn and mimic
user actions in applications
or systems

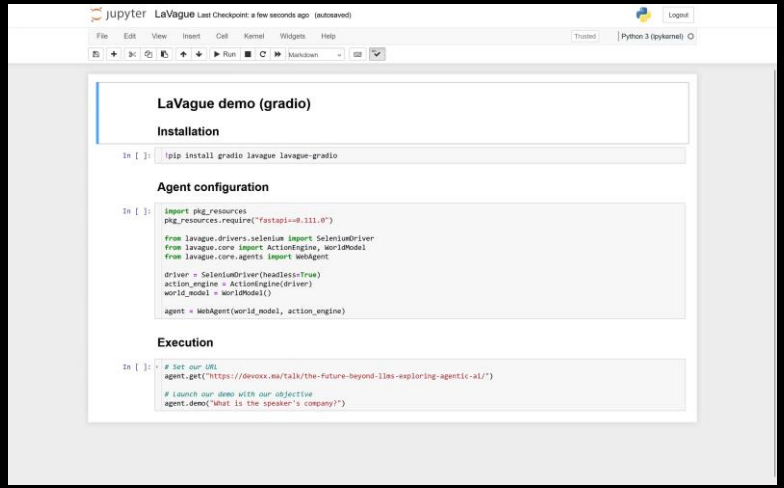
WORLDLINE

Large Action Models - LaVague



Large Action Model framework for developing AI Web Agents

- Open source framework to automate automation
- Trained of the BigAction dataset
  - Pairs of (query, HTML) -> (Selenium code)
  - Generates the next instruction to achieve goal
  - Compiles it into action code (Selenium)
- Usages: Code, Gradio app or Chrome extension
- Use cases
  - Retrieve knowledge (ex. extract information)
  - Data Entry (ex. medical appointment booking)
  - Web test & QA
- Demo: extract information from a web page



Large Action Models – Embodied AI

DEVOXX MOROCCO

LAMs enabling physical devices
to understand and act upon the world

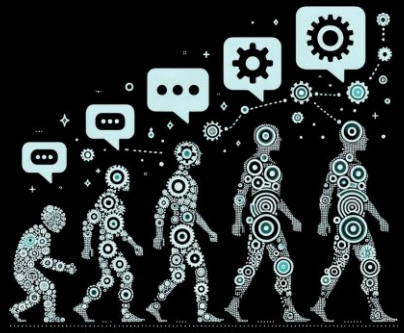
- **New type of devices: Rabbit R1**
  - Autonomous wearable
  - Natural Language Interface
  - Connected to real-world applications
  - No applications on the device but a LAM-powered OS



- **Use in Robotics: Reachy 2**
  - By Pollen Robotics/Hugging Face
  - Trained on videos in VR of household chores
  - Each video correspond to a specific sensor
  - Model and dataset are open source



Wrap-up



Conversational AI → Agentic AI → Embodied AI

- **AI Revolution?**
  - **Beyond Chatbots**: Conversational AI is evolving, becoming smarter, capable of reasoning and planning
  - **Actionable AI**: Multi-agent systems and LAMs are bringing AI into the real world, driving complex workflows
- **A Future of Synergy**
  - **Building on Success:** AI is a layered science, with each innovation building on the last
  - **Unleashing Potential**: Let's combine LLMs, Agents, and LAMs for truly powerful AI

Raphaël Semeteys
raphiki.github.io



Thanks You

DEVOXX™ MOROCCO

WORLDLINE